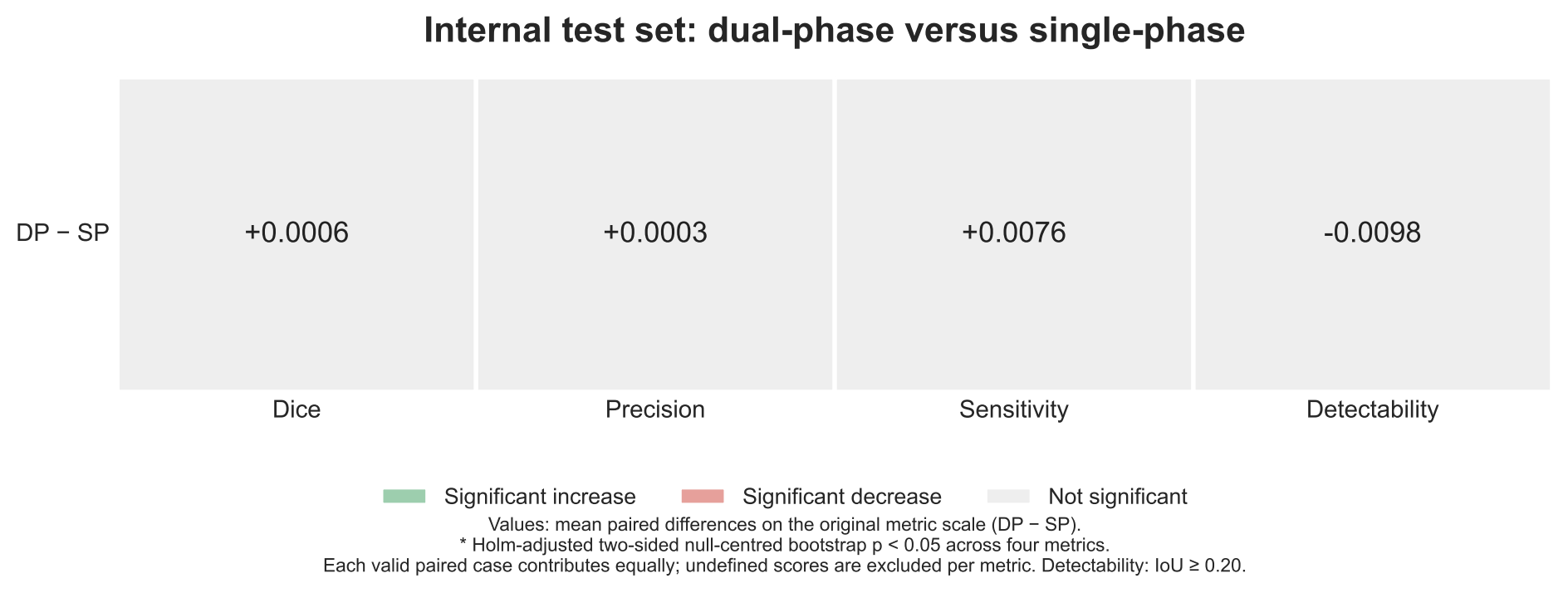
Internal test set: dual-phase versus single-phase
DP − SP
+0.0006
+0.0003
+0.0076
-0.0098
Dice
Precision
Sensitivity
Detectability
Significant increase
Significant decrease
Not significant
Values: mean paired differences on the original metric scale (DP − SP).
* Holm-adjusted two-sided null-centred bootstrap p < 0.05 across four metrics.
Each valid paired case contributes equally; undefined scores are excluded per metric. Detectability: IoU ≥ 0.20.